

Polynomial Interpolation

1.2 Newton's Divided Difference

One challenge in using Lagrange interpolation is that when constructing a higher-order interpolation polynomial, we cannot directly reuse a lower-order polynomial that has already been computed. Specifically, if we construct an interpolation polynomial $P_1(x)$ using the data points (x_0, y_0) and (x_1, y_1) , we define the corresponding basis functions

$$l_0(x) = \frac{x - x_1}{x_0 - x_1}, \quad l_1(x) = \frac{x - x_0}{x_1 - x_0}.$$

However, when adding a new data point (x_2, y_2) and constructing a higher-order interpolation polynomial $P_2(x)$, the previously computed basis functions cannot be reused as they are. Instead, all basis functions

$$l_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}, \quad l_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}, \quad l_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

must be recalculated from scratch.

This inefficiency makes Lagrange interpolation computationally expensive, especially when dealing with large datasets or when the interpolation order needs to be increased iteratively. To address this limitation, the **divided differences** method provides a more structured and recursive way to compute interpolation polynomials.

Case 1: Linear Polynomial Interpolation

Find a linear polynomial interpolating a function $f(x)$ at (x_0, y_0) and (x_1, y_1) where $f(x_0) = y_0$, $f(x_1) = y_1$.

Case 2: Quadratic Polynomial Interpolation

Consider the construction of a polynomial of degree at most 2 that passes through three points $(x_0, y_0), (x_1, y_1), (x_2, y_2)$.

Then $P_2(x) =$

We need $P_2(x)$ to agree with the function f at x_0, x_1, x_2 . i.e.,

$$P_2(x_0) = f(x_0) \implies$$

$$P_2(x_1) = f(x_1) \implies$$

$$P_2(x_2) = f(x_2) \implies$$

The generation of the divided differences is outlined in the following table.

x	$f(x)$	First divided differences	Second divided differences	Third divided differences
x_0	$f[x_0]$	$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$	$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$	$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$	$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	
x_3	$f[x_3]$			

Example 1. Find the linear polynomial passing through the points $(0, 3)$, $(1, 0)$, $(2, 2)$ using the divided differences.

Case 3: General Case

Theorem 2

Suppose that $P_n(x)$ is the n -th degree interpolating polynomial that agrees with the function f at the distinct numbers $x_0, x_1, \dots, x_n$. The divided differences of f with respect to $x_0, x_1, \dots, x_n$ are used to express $P_n(x)$ in the form:

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0)(x - x_1) \dots (x - x_{n-1}),$$

for appropriate constants $a_0, a_1, \dots, a_n$. In particular,

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \dots (x - x_{n-1}),$$

where the divided differences are defined as follows:

Zero-th divided difference (w.r.t. x_i):

$$f[x_i] = f(x_i)$$

First divided difference (w.r.t. x_i, x_{i+1}):

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

Second divided difference (w.r.t. x_i, x_{i+1}, x_{i+2}):

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

k -th divided difference (w.r.t. $x_i, x_{i+1}, \dots, x_{i+k}$):

$$f[x_i, x_{i+1}, \dots, x_{i+k}] = \frac{f[x_{i+1}, x_{i+2}, \dots, x_{i+k}] - f[x_i, x_{i+1}, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$